

AI Knowledge Retrieval & Analytics Assistant

Objective:

Create a Secure RAG-based AI assistant that indexes terabytes of enterprise data(PDFs, docs, databases, logs) and answers natural language questions with citations and analytics.

Architecture:

1. Data Sources

- SQL databases, CSVs, PDFs, APIs
- Ingested via Spark ETL pipelines -> stored in data lake (s3/GCS)

2. Embedding + Indexing

- Use OpenAI/ SentenceTransformers to generate Embeddings
- Store in **FAISS** / **Pinecone** / **Elasticsearch** Vector DB

3. Retrieval-Augmented Generation(RAG)

- Query pipeline
 User Query -> Embedding Match -> Context Fetch -> LLM Response
- Use LangChain / LlamaIndex to manage context windows

4. LLM Backend

- Open-source model (Llama 3, Mistral) fine-tuned on internal style
- Deployed via Vertex AI / AWS sagemaker

5. Analytics Layer

- Spark for log aggregation and query analytics
- Generates insight report like “Most queried topics this month”

6. Access Control + API's

- FastAPI + OAuth2 for Authentication
- Multi-tenant design with usage tracking per Organisation

Tech Stack:

ML/AI: LangChain, SentenceTransformers, PyTorch

Data: Spark, SQL, Pandas

Infra: GCP Vertex AI / AWS SageMaker

Vector DB: Pinecone / FAISS

Frontend: Streamlit / React dashboard

